**Questions:**

**IN & BETWEEN**

1. List the type of the courses “Statistics” and “Programming”
2. The instructor wants to know the CourseNos whose scores are in the range 50 to 80

**AGGREGATE**

1. Find the average mark of “C002”.
2. List the maximum, minimum mark for “C002”
3. List the maximum, minimum, average mark for each course
4. List the name of the courses and average mark of each course.
5. Calculate the sum of all the scores.
6. How many students are registered for each course? Display the course description and the number of students registered in each course.
7. How many courses did each student register for?

```
mysql> USE CollegeDB;
Database changed
mysql>
mysql> -- CREATE MARKS TABLE
mysql> CREATE TABLE Marks
    -> (
    ->      RegNo INT,
    ->      CourseID VARCHAR(10),
    ->      Score INT,
    ->      FOREIGN KEY (RegNo) REFERENCES Student(RegNo),
    ->      FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql>
mysql> -- INSERT SAMPLE DATA INTO MARKS
mysql> INSERT INTO Marks VALUES
    -> (101,'C001',75),
    -> (101,'C002',68),
    -> (102,'C002',82),
    -> (103,'C001',55),
    -> (104,'C003',90),
    -> (105,'C001',65),
    -> (106,'C004',72),
    -> (107,'C002',78),
    -> (108,'C003',88);
Query OK, 9 rows affected (0.01 sec)
Records: 9  Duplicates: 0  Warnings: 0

mysql>
mysql> -- ADD NEW COURSES
mysql> INSERT INTO Course VALUES
    -> ('C005','Statistics'),
    -> ('C006','Programming');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> -----------------------------------------------------------
    -> -- IN & BETWEEN
    -> -----------------------------------------------------------
    ->
    -> -- 1. List the CourseID and CourseName for Statistics and Pro
    -> SELECT CourseID, CourseName
    -> FROM Course
    -> WHERE CourseName IN ('Statistics','Programming');
ERROR 1064 (42000): You have an error in your SQL syntax; check the
-----------------------------------------------------

-----------------------' at line 1
mysql>
mysql> -- 2. List the CourseIDs whose scores are between 50 and 80
mysql> SELECT CourseID
    -> FROM Marks
    -> WHERE Score BETWEEN 50 AND 80;
+----------+
| CourseID |
+----------+
| C001     |
| C002     |
| C001     |
| C001     |
| C004     |
| C002     |
+----------+
6 rows in set (0.00 sec)
```

```
mysql> -- 2. Find the maximum and minimum mark for C002
mysql> SELECT MAX(Score) AS Maximum_Mark,
    ->        MIN(Score) AS Minimum_Mark
    -> FROM Marks
    -> WHERE CourseID='C002';
+--------------+--------------+
| Maximum_Mark | Minimum_Mark |
+--------------+--------------+
|           82 |           68 |
+--------------+--------------+
1 row in set (0.00 sec)

mysql>
mysql> -- 3. Find the maximum, minimum and average mark for each course
mysql> SELECT CourseID,
    ->        MAX(Score) AS Maximum_Mark,
    ->        MIN(Score) AS Minimum_Mark,
    ->        AVG(Score) AS Average_Mark
    -> FROM Marks
    -> GROUP BY CourseID;
+----------+--------------+--------------+--------------+
| CourseID | Maximum_Mark | Minimum_Mark | Average_Mark |
+----------+--------------+--------------+--------------+
| C001     |           75 |           55 |      65.0000 |
| C002     |           82 |           68 |      76.0000 |
| C003     |           90 |           88 |      89.0000 |
| C004     |           72 |           72 |      72.0000 |
+----------+--------------+--------------+--------------+
4 rows in set (0.00 sec)

mysql>
mysql> -- 4. Display the course name and average mark of each course
mysql> SELECT C.CourseName,
    ->        AVG(M.Score) AS Average_Mark
    -> FROM Course C
    -> JOIN Marks M
    -> ON C.CourseID = M.CourseID
    -> GROUP BY C.CourseName;
```

```
    -> GROUP BY C.CourseName;
+--------------------+---------------+
| CourseName         | Average_Mark |
+--------------------+---------------+
| DBMS               |      65.0000 |
| Java               |      76.0000 |
| Python             |      89.0000 |
| Operating System   |      72.0000 |
+--------------------+---------------+
4 rows in set (0.01 sec)

mysql>
mysql> -- 5. Calculate the sum of all scores
mysql> SELECT SUM(Score) AS Total_Score
    -> FROM Marks;
+-------------+
| Total_Score |
+-------------+
|         673 |
+-------------+
1 row in set (0.00 sec)

mysql>
mysql> -- 6. Display the course description and the number of students registered in each course
mysql> SELECT C.CourseName,
    ->        COUNT(R.RegNo) AS Number_of_Students
    -> FROM Course C
    -> JOIN Registration R
    -> ON C.CourseID = R.CourseID
    -> GROUP BY C.CourseName;
+--------------------+--------------------+
| CourseName         | Number_of_Students |
+--------------------+--------------------+
| DBMS               |                  3 |
| Java               |                  2 |
| Python             |                  2 |
| Operating System   |                  1 |
+--------------------+--------------------+
4 rows in set (0.00 sec)
```

```
4 rows in set (0.00 sec)

mysql>
mysql> -- 7. Display how many courses each student registered for
mysql> SELECT S.RegNo,
    ->        S.Name,
    ->        COUNT(R.CourseID) AS Total_Courses
    -> FROM Student S
    -> JOIN Registration R
    -> ON S.RegNo = R.RegNo
    -> GROUP BY S.RegNo, S.Name;
+-------+--------+---------------+
| RegNo | Name   | Total_Courses |
+-------+--------+---------------+
|   101 | Ramesh |             1 |
|   102 | Suresh |             1 |
|   103 | Kamesh |             1 |
|   104 | Priya  |             1 |
|   105 | Shyam  |             1 |
|   106 | Ramya  |             1 |
|   107 | Aakash |             1 |
|   108 | Amala  |             1 |
+-------+--------+---------------+
8 rows in set (0.00 sec)
```